

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE PROJECT

Final Project Report

Submitted By:

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Tools & Technologies:

Python, Pandas, SQLite, SQL, Jupyter Notebook, Power BI, Git, GitHub

Date:

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EXECUTIVE SUMMARY

This project focuses on analyzing mutual fund industry data using Python, SQLite, SQL, Jupyter Notebook, and Power BI. The project includes data ingestion, cleaning, transformation, exploratory data analysis, performance analytics, advanced risk analysis, and dashboard development. The objective is to generate meaningful insights from mutual fund and investor datasets and present them through interactive visualizations and reports.

PROJECT OBJECTIVES

- Build an end-to-end ETL pipeline.
- Clean and validate mutual fund datasets.
- Store processed data in SQLite.
- Perform exploratory data analysis.
- Compute performance metrics such as Sharpe Ratio, Alpha, Beta, CAGR, and Drawdown.
- Perform advanced analytics including VaR, CVaR, and Investor Cohort Analysis.
- Build an interactive Power BI dashboard.
- Generate actionable business insights.

DATA SOURCES

1. Fund Master Data
2. NAV History Data
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category Inflows
6. Industry Folio Counts
7. Scheme Performance Data
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

ETL ARCHITECTURE

Raw CSV Files



Python ETL Scripts



Data Cleaning & Validation



Processed CSV Files



SQLite Database



EDA & Performance Analytics



Power BI Dashboard

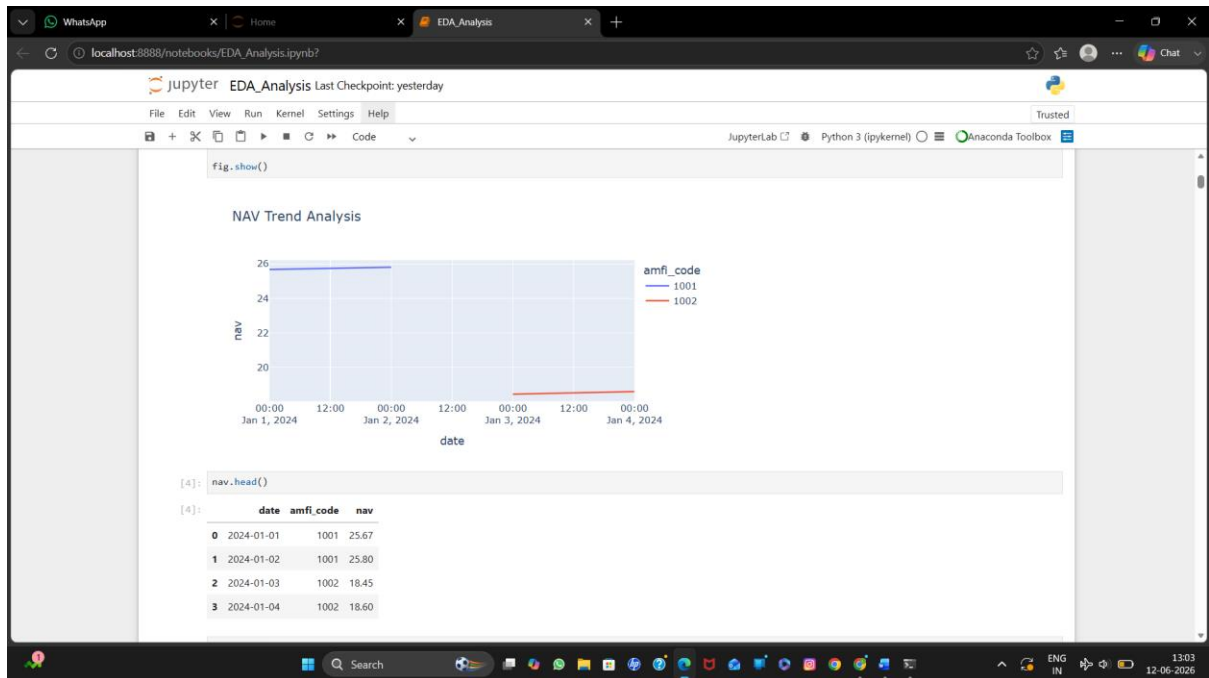


Final Reports & Insights

DATA CLEANING & VALIDATION

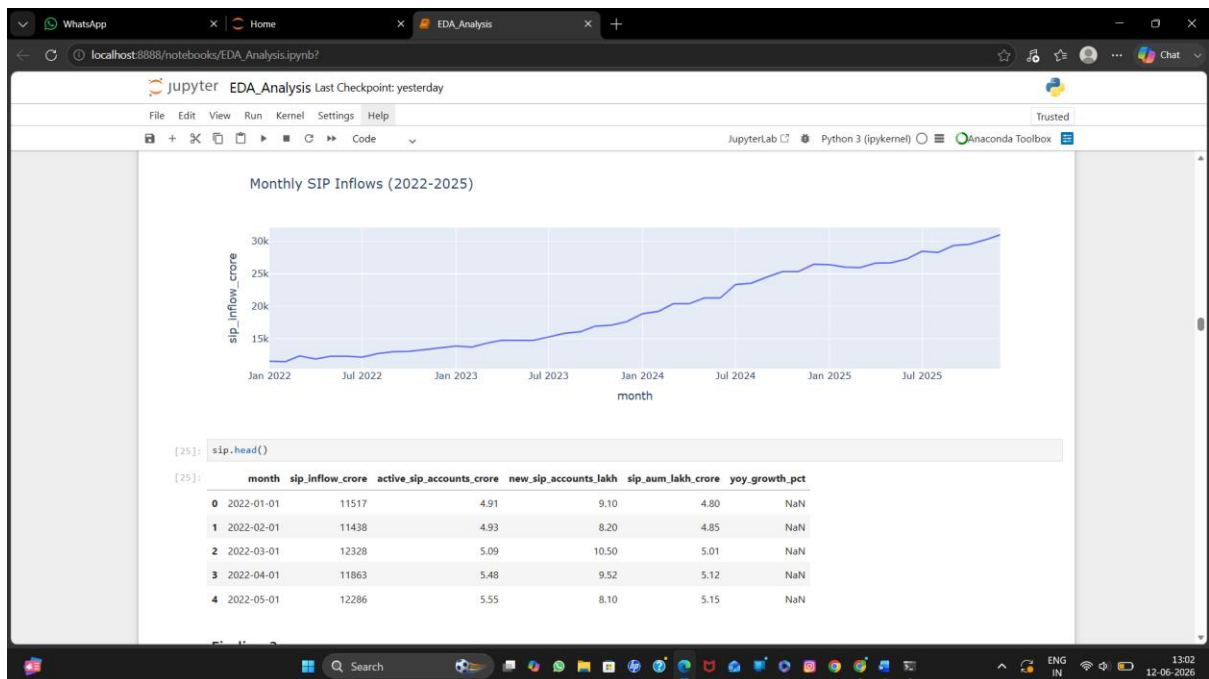
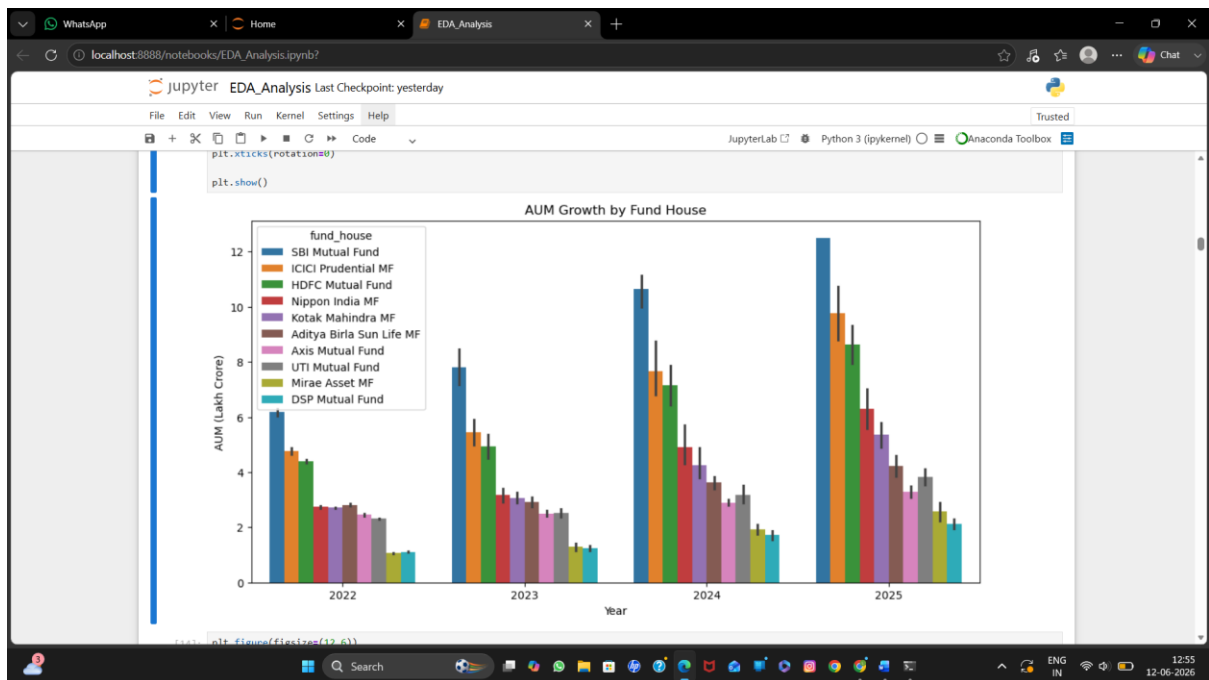
The raw datasets were cleaned and validated using Python and Pandas. Missing values, duplicate records, invalid dates, and inconsistent formats were identified and corrected. Transaction records were standardized, NAV values were validated, and data quality checks were performed before loading the processed data into SQLite.

NAV TREND ANALYSIS



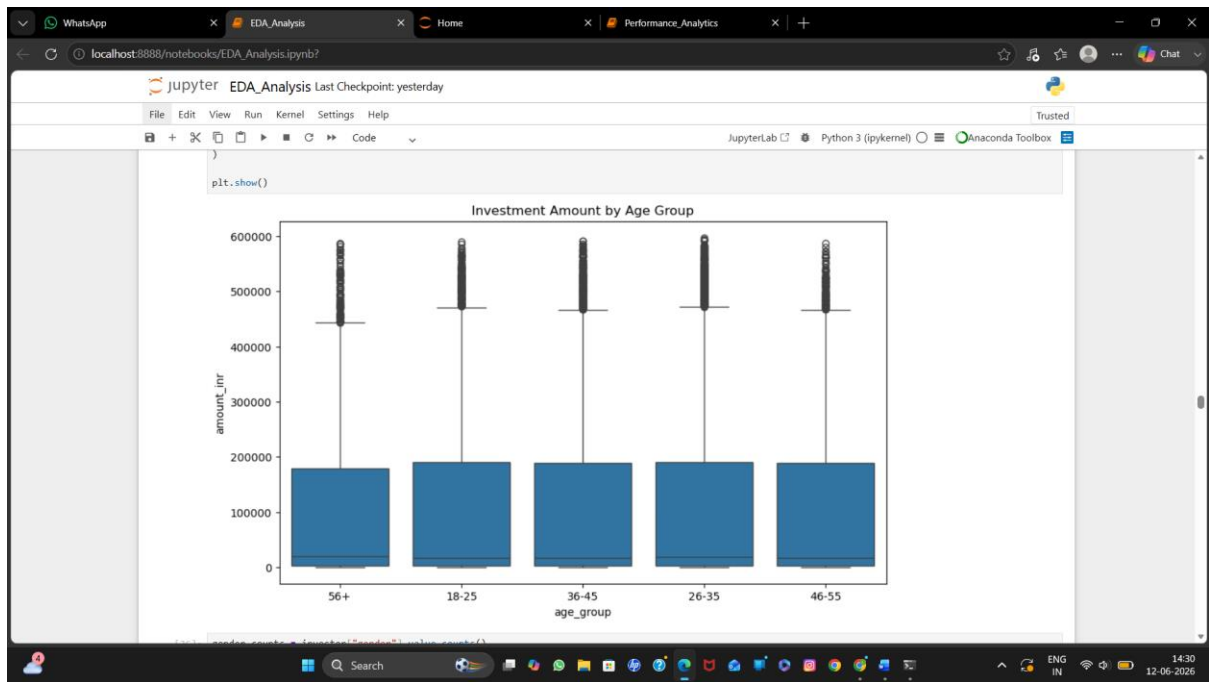
The NAV trend analysis shows a steady growth pattern across multiple mutual fund schemes between 2022 and 2025. Most equity-oriented funds demonstrated positive long-term appreciation despite short-term market fluctuations.

INDUSTRY & SIP ANALYSIS



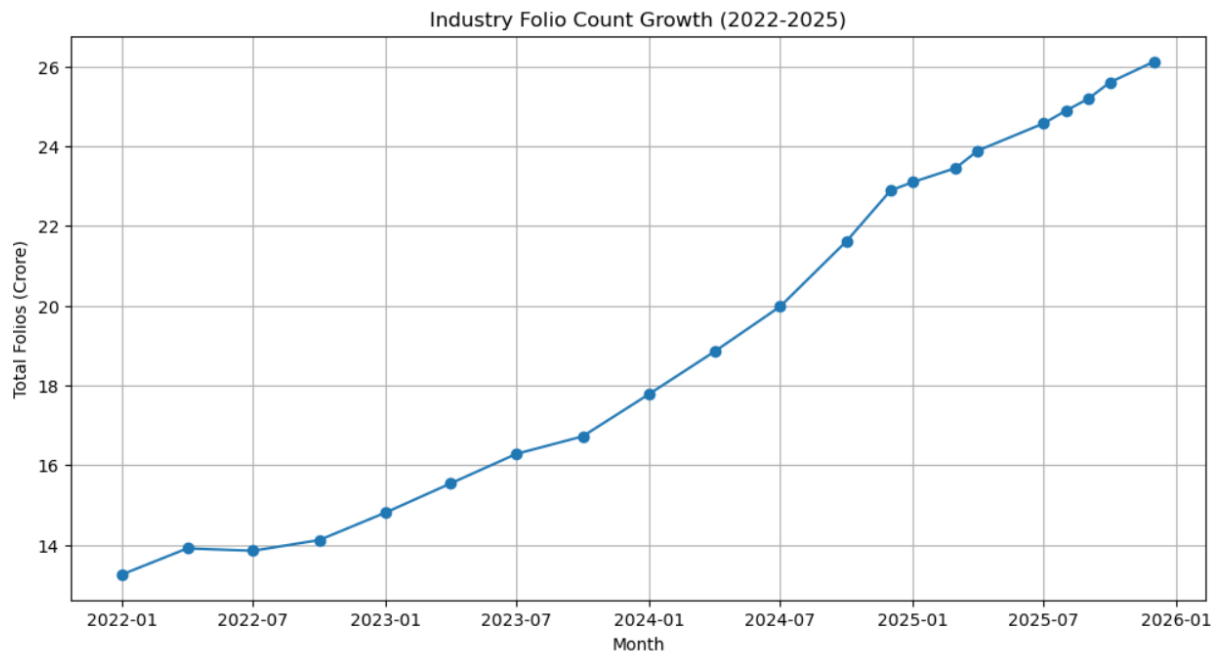
Industry assets under management (AUM) increased significantly from 2022 to 2025. Monthly SIP inflows also showed consistent growth, reflecting increasing investor participation and confidence in mutual funds.

CATEGORY & INVESTOR ANALYSIS



Equity-oriented categories attracted the highest inflows. Investor participation was concentrated in specific age groups, with SIP investments contributing the largest share of retail investments.

FOLIO GROWTH ANALYSIS



Total mutual fund folios increased substantially during the study period, indicating strong growth in investor adoption and market penetration.

PERFORMANCE ANALYTICS OVERVIEW

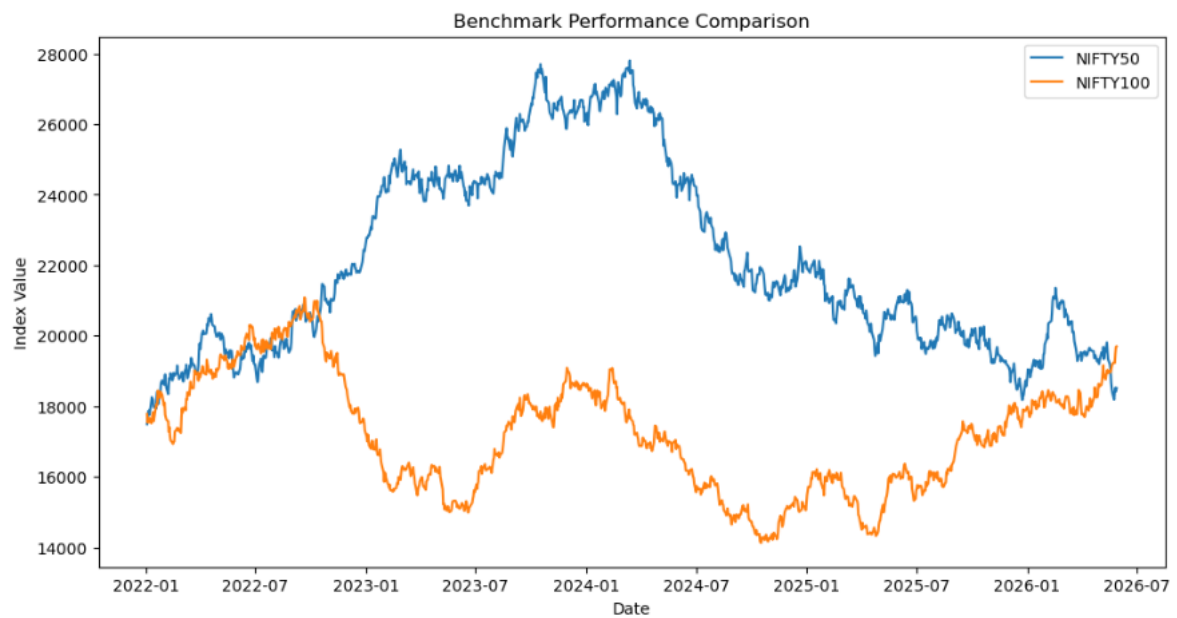
Performance analytics were conducted using CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Benchmark Comparison metrics. These measures were used to evaluate return generation, risk-adjusted performance, and downside protection across mutual fund schemes.

FUND SCORECARD & RANKINGS

	scheme_name	fund_score
3	SBI Small Cap Fund - Direct Plan - Growth	72.75
22	Kotak Flexicap Fund - Regular - Growth	71.50
21	Kotak Emerging Equity Fund - Regular - Growth	70.50
29	ABSL Small Cap Fund - Regular - Growth	68.25
2	SBI Small Cap Fund - Regular Plan - Growth	64.38

The fund scorecard ranked schemes using return, Sharpe ratio, alpha, expense ratio, and drawdown metrics to identify the strongest performers.

BENCHMARK COMPARISON



Fund performance was compared against market benchmarks such as NIFTY50 and NIFTY100. Top-performing funds consistently outperformed benchmark returns over the evaluation period.

ADVANCED ANALYTICS

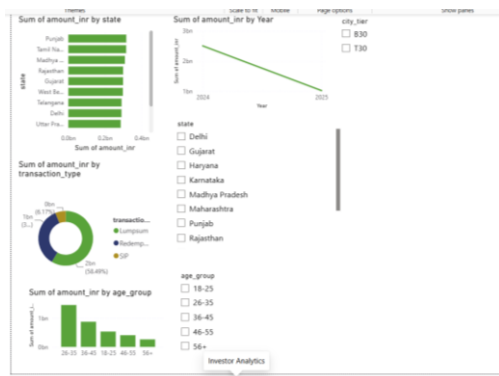
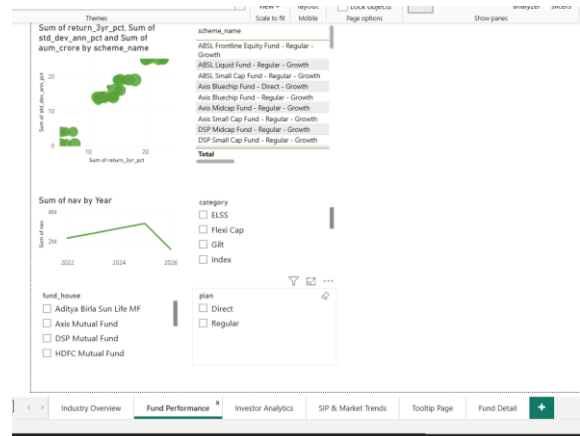
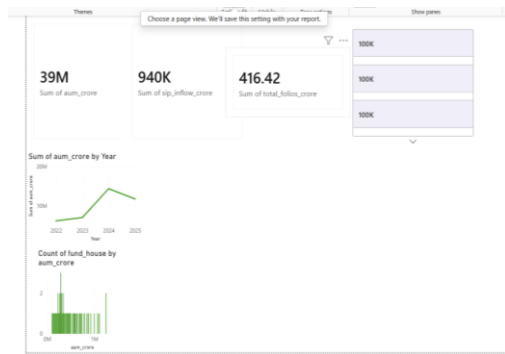
Advanced analytics included Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, and Portfolio Concentration Measurement using the Herfindahl-Hirschman Index (HHI).

ADVANCED INSIGHTS

1. High-risk funds exhibited larger VaR and CVaR values.
2. Long-term investor cohorts invested significantly more than newer cohorts.

3. Most SIP investors maintained regular contribution schedules.
4. Diversified funds showed lower concentration risk than focused sector funds.
5. Funds with strong Sharpe Ratios generally ranked highest in the recommendation engine.

POWER BI DASHBOARD OVERVIEW



An interactive Power BI dashboard was developed to provide real-time exploration of mutual fund performance, investor behavior, SIP trends, and industry-level insights.

Limitations

- Dataset is limited to the provided sample data.
- No real-time market data integration.
- Simplified recommendation engine.
- Historical analysis only.

Recommendations

- Integrate live NAV APIs.
- Add predictive analytics models.
- Deploy dashboard to cloud platforms.
- Expand recommendation logic using machine learning.

Conclusion

The Bluestock Mutual Fund Analytics Capstone successfully implemented an end-to-end analytics solution covering ETL, data warehousing, exploratory analysis, risk analytics, performance measurement, and business intelligence reporting. The project demonstrates practical application of data analytics techniques to solve real-world financial analysis problems and generate actionable insights.